Online PES Decoder Learning in Neuromorphic Residual Control of UAV Quaternion Tracking

Gandhi Contreras[1], Steven Sandoval[1], and Luis Rodolfo Garcia Carrillo[2]

***Abstract*— Online neuromorphic residual control can augment a classical attitude controller under abrupt missions or unforeseen disturbances without changing its feedback gains or actuator interface. We present six continuous coupled membrane-gate (CMG) units and eighteen decoder weights adapted by prescribed error sensitivity (PES), adding a bounded correction to the saturated quaternion proportional-derivative (PD) torque request before rotor-force allocation. Across ten matched 45-second MuJoCo cases on the Skydio X2 and a Crazyflie 2 derived model, mean quaternion-angle error decreases by 17.3–60.6% relative to that fixed PD law. Because the law damps absolute body rate rather than the rate error its own kinematics define, we repeat all ten cases on a reference-rate feedforward PD and on a bandwidth-tuned one: the residual still reduces error, by 57.8/57.6% and by 35.6/6.5% in geometric mean, while the tuned classical law on its own outperforms the PD-referenced augmented controller in every cell, and on the abrupt step mission the residual degrades tracking. A five-family neuron benchmark exposes an effort-accuracy tradeoff: CMG–PES matches the strongest learned basis at a third of its residual effort, and decoder adaptation, not neuron dynamics, separates the improving cases. Under synthetic measurement noise the reductions persist, and learned corrections are reused across repeated disturbances. The residual adds bounded, revocable adaptation whose benefit is largest where the baseline lacks a model term.**

I. Introduction

Learning-based augmentation can retain classical feedback while adapting a correction that is difficult to specify *a priori*. We study a bounded torque residual for quaternion attitude tracking of an unmanned aerial vehicle (UAV) with individual rotor-force saturation: whether a small stateful basis improves a fixed PD controller, whether the improvement comes from neuron dynamics, decoder learning, or correction magnitude, and how it survives a stronger classical baseline. Bounded authority makes the learning layer switchable; it does not by itself ensure stability or safety.

Prior work includes residual robot learning [1], geometric $\mathcal{L}_1$ quadrotor augmentation with performance guarantees [2], and proportional–integral–derivative (PID) attitude control augmented with a proximal policy optimization (PPO) residual [3]. Brain-inspired robust UAS controllers likewise pair model-based feedback with learned terms, from limbic-system designs [4], [5] to neural weight reset under impulsive disturbances [6], and perturbation estimation targets the same disturbances on small quadrotors [7]. Neuromorphic control offers a complementary implementation approach: Nengo supports the Neural Engineering Framework (NEF) [8] and online decoder learning [9], most often through the prescribed error sensitivity (PES) rule [10], an established local, online error-driven update [11]. PES-based adaptive control has been demonstrated on manipulators [12], [13], and in self-tuning spiking PID controllers for free-flying UAS trajectory tracking [14], [15]. Building on that line, our focus is a continuous membrane-gate basis for a bounded quaternion-torque residual, with a fixed classical controller and adaptation confined to its decoder.

We augment a quaternion PD baseline with a compact neuromorphic residual architecture based on coupled conductance-based membrane-gate (CMG) units and PES decoder learning. The paper's contributions are:

1) a code-aligned CMG–PES architecture with paired currents, explicit encoding, state-update, and decoder/readout equations, and a fully specified actuator interface (Fig. 1);
2) proofs of internal-state, command-deviation, and weight boundedness, with exact recovery of the PD baseline at zero residual authority, and an allocation-aware local input-to-state certificate whose constants are computed for both vehicles;
3) a controlled ablation that separates conductance modulation, decoder learning, and output authority, and identifies which mechanism is primarily responsible for the tracking-error reduction; and
4) robustness evidence that the reduction survives realistic measurement noise, together with a hold-equilibrium study in which the learned correction is reused across repeated disturbance episodes rather than relearned; and
5) a baseline ladder that re-evaluates the residual against a reference-rate feedforward PD and a bandwidth-tuned one, reporting both the reductions it retains and the cells in which the classical law wins.

The CMG unit couples a leaky membrane to a gating state that modulates its effective conductance, linking LIF-like membrane integration with Hodgkin–Huxley inspired gating dynamics [16]. Although related operating regimes can support spike-like responses, the present controller deliberately exploits the continuous, non-spiking membrane-gate trajectory, which provides a compact set of stateful temporal features for the residual. The present implementation is digital; the same temporal-feature mechanism holds regardless of computing platform, and no neuromorphic-hardware energy advantage is claimed.

[1]New Mexico State University (NMSU), Las Cruces, NM, USA.

[2] Space Vehicles Directorate, Air Force Research Laboratory (AFRL), Kirtland Air Force Base (KAFB), Albuquerque, NM 87108. Email: luis_rodolfo.garcia_carrillo@afrl.af.mil

Section II defines the problem and actuator interface; Section III presents the CMG–PES architecture, its boundedness properties, and a local stability certificate; Section IV reports the simulation study, ablations, benchmarks, and noise and disturbance experiments; and Section V discusses limitations and future work. Code, model files, and run archives will be available at `https://github.com/[to-be-released]`.

II. Attitude Baseline and Actuator Interface

The objective is to reduce shortest-path attitude error for a prescribed $q_d(t)$ while respecting rotor-force limits. Baseline and augmented controllers share the state measurements, height loop, and allocator; Section III modifies only the requested attitude torque.

A. Quaternion Convention and Error

Let $Q \in \mathbb{S}^3$ denote the scalar-first Hamilton quaternion representing attitude (body to inertial), and let $\boldsymbol{\omega}$ be the body angular velocity. For a rigid body with inertia $\boldsymbol{J} \succ 0$ about the center of mass, expressed in body axes (cf. Fig. 1), the free-flight rotational dynamics are [17]

$$\dot{Q} = \tfrac{1}{2} Q \otimes [0, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T}, \tag{1}$$

$$\boldsymbol{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\mathrm{act}} + \boldsymbol{\tau}_{\mathrm{p}} + \boldsymbol{\tau}_{\mathrm{d}} - \boldsymbol{\omega} \times (\boldsymbol{J}\boldsymbol{\omega}). \tag{2}$$

Here $\boldsymbol{\tau}_{\mathrm{p}}$ denotes modeled passive torque and $\boldsymbol{\tau}_{\mathrm{d}}$ any applied disturbance; the nominal ideal rigid-body model sets both to zero. Contact, if present, contributes an additional external wrench.

We work with the corresponding passive quaternion $q = Q^{-1}$ [18], which satisfies

$$\dot{q} = -\tfrac{1}{2}[0, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T} \otimes q. \tag{3}$$

The desired attitude q_d uses the same convention. Define the attitude error

$$q_e = q_d \otimes q^{-1} = [q_{e,0}, \boldsymbol{q}_{e,v}^\mathsf{T}]^\mathsf{T}, \tag{4}$$

$$\boldsymbol{e}_q = \mathrm{sgn}_+(q_{e,0})\boldsymbol{q}_{e,v}, \qquad \mathrm{sgn}_+(0) = 1, \tag{5}$$

$$\theta_q = 2\,\mathrm{atan2}(\|\boldsymbol{q}_{e,v}\|_2, |q_{e,0}|) \in [0, \pi]. \tag{6}$$

The reported error θ_q is the shortest rotation angle, invariant under $q_e \mapsto -q_e$. Sign selection introduces a discontinuity at $q_{e,0} = 0$; no continuous feedback globally stabilizes rigid-body attitude [19], [20], robust global tracking requires hybrid [21] or geometric-algebra [22], [23] formulations, and the analysis here is local.

For a desired-frame rate $\boldsymbol{\omega}_d$ satisfying the corresponding passive kinematics, differentiation gives

$$\dot{q}_e = \tfrac{1}{2} q_e \otimes [0, \boldsymbol{\omega}_e^\mathsf{T}]^\mathsf{T}, \tag{7}$$

$$\boldsymbol{\omega}_e = \boldsymbol{\omega} - R(q_e)^\mathsf{T}\boldsymbol{\omega}_d, \qquad \dot{q}_{e,0} = -\tfrac{1}{2}\boldsymbol{q}_{e,v}^\mathsf{T}\boldsymbol{\omega}_e. \tag{8}$$

These are kinematic identities (no convergence implied). Reference Euler triples parameterize q_d in this passive convention.

B. Physical Inertia and Implemented PD Gains

When principal moments $\boldsymbol{j}_{\mathrm{p}}$ of $\boldsymbol{J}$ are specified in a rotated principal-axis frame, the physical body-frame tensor is

$$\boldsymbol{J} = R_{BI}\,\mathrm{diag}(\boldsymbol{j}_{\mathrm{p}})R_{BI}^\mathsf{T}, \tag{9}$$

for the rotation from principal axes to body axes R_{BI}. The implementation sets axiswise gains using the principal moments in their reported ordering, $\boldsymbol{j}_{\mathrm{tune}}$, not necessarily $\mathrm{diag}(\boldsymbol{J})$ (Table I).

The fixed quaternion PD baseline is

$$\boldsymbol{k}_q = 2\boldsymbol{j}_{\mathrm{tune}} \odot \boldsymbol{\omega}_n^2, \qquad \boldsymbol{k}_\omega = 2\zeta\boldsymbol{j}_{\mathrm{tune}} \odot \boldsymbol{\omega}_n, \tag{10}$$

$$\boldsymbol{u}_0 = \mathrm{sat}_{\boldsymbol{\ell}}(-\boldsymbol{k}_q \odot \boldsymbol{e}_q - \boldsymbol{k}_\omega \odot \boldsymbol{\omega}), \tag{11}$$

where $\odot$ denotes componentwise multiplication, $\boldsymbol{\omega}_n = (4, 4, 3.5)^\mathsf{T}$ rad/s, $\zeta = 0.95$, and $\mathrm{sat}_{\boldsymbol{\ell}}$ clips component i to $[-\ell_i, \ell_i]$. The requested-torque box is $\mathcal{U} = \prod_{i=1}^{3}[-\ell_i, \ell_i]$; $\mathrm{clip}_{[a,b]}$ also acts componentwise, and $\oslash$ denotes componentwise division. We impose componentwise torque limits $\boldsymbol{\ell} \in \mathbb{R}^3_{>0}$ through $\mathrm{sat}_{\boldsymbol{\ell}}(\cdot)$. In our implementation, $\boldsymbol{\ell}$ is chosen proportional to the diagonal gain-scheduling vector $\boldsymbol{j}_{\mathrm{tune}}$ via $\ell_i = \max(j_{\mathrm{tune},i}\nu_i, 10^{-6}\ \mathrm{N\,m})$, with acceleration-scale vector $\boldsymbol{\nu} = (12, 12, 10)^\mathsf{T}$ rad/s^2. These controller-side bounds are not the rotor-feasible torque set. The law is quaternion feedback [24]; $\boldsymbol{\omega}_n$ is a tuning parameter rather than a claimed set of coupled closed-loop natural frequencies. Equation (11) damps the absolute rate, while (8) defines the rate error $\boldsymbol{\omega}_e$. Because that gap, not the neural layer, dominates moving-reference tracking, we also report the feedforward baseline

$$\boldsymbol{u}_0^{\mathrm{ff}} = \mathrm{sat}_{\boldsymbol{\ell}}\big(-\boldsymbol{k}_q \odot \boldsymbol{e}_q - \boldsymbol{k}_\omega \odot (\boldsymbol{\omega} - \boldsymbol{\omega}_d)\big), \tag{12}$$

with $\boldsymbol{\omega}_d$ from central differences of the reference, and its bandwidth-scaled variant $\boldsymbol{\omega}_n \mapsto \rho\boldsymbol{\omega}_n$ (Sec. IV-C). Gyroscopic cancellation remains omitted. The resulting gains are vehicle-specific through $\boldsymbol{j}_{\mathrm{tune}}$, even though $\boldsymbol{\omega}_n$ and ζ are shared; the same fixed gains are then retained for PD and its CMG augmentation. The local result in Section III-D uses these anisotropic gains; it does not replace them with a scalar-gain controller.

C. Requested and Realized Wrench

For rotor positions $\boldsymbol{p}_i$ in body coordinates, parallel $+z$ thrust axes, and signed reaction coefficients κ_i, define

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ p_{1,y} & p_{2,y} & p_{3,y} & p_{4,y} \\ -p_{1,x} & -p_{2,x} & -p_{3,x} & -p_{4,x} \\ \kappa_1 & \kappa_2 & \kappa_3 & \kappa_4 \end{bmatrix}. \tag{13}$$

The signs follow $\boldsymbol{p}_i \times [0, 0, f_i]^\mathsf{T}$ and the actuator's local reaction torque $\kappa_i f_i$ [25]. Here B maps rotor thrusts to total thrust and body torque; κ_i has units of meters, not angular velocity. For four rotors B^{-1} equals the Moore–Penrose inverse $B^\dagger$ that other rotor counts use. Given a commanded collective thrust T and body-torque command

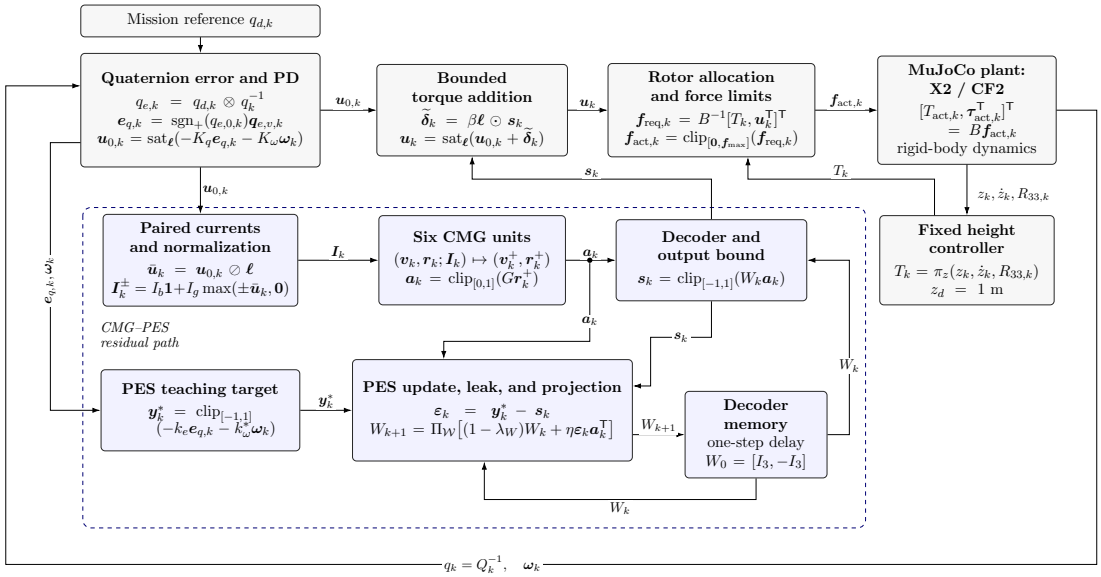


Fig. 1. Online PES decoder learning CMG-PES Neuromorphic Residual Control architecture

$\boldsymbol{u}$, the corresponding inverse-allocation and clipping model is

$$\begin{aligned} \boldsymbol{f}_{\mathrm{req}} &= B^{-1}[T, \boldsymbol{u}^\mathsf{T}]^\mathsf{T}, \\ \boldsymbol{f}_{\mathrm{act}} &= \mathrm{clip}_{[\mathbf{0}, \boldsymbol{f}_{\max}]}(\boldsymbol{f}_{\mathrm{req}}), \\ [T_{\mathrm{act}}, \boldsymbol{\tau}_{\mathrm{act}}^\mathsf{T}]^\mathsf{T} &= B \boldsymbol{f}_{\mathrm{act}}. \end{aligned} \tag{14}$$

Here $\boldsymbol{u} = \boldsymbol{u}_0$ for PD; the CMG path in Section III modifies that requested torque before the same allocation and clipping. Each f_i is a commanded force, not a rotor-speed command, and exact realization requires $B^{-1}[T, \boldsymbol{u}^\mathsf{T}]^\mathsf{T} \in [\mathbf{0}, \boldsymbol{f}_{\max}]$: a componentwise torque box is not generally inside this collective-dependent set. Allocation is clipped inverse, not optimization-based [26].

Collective thrust remains classical, with $z_d = 1$ m:

$$T = \mathrm{clip}_{[0, 2.5Mg]} \left(\frac{M[g + 6(z_d - z) - 4\dot{z}]}{\max(R(Q)_{33}, 0.35)} \right). \tag{15}$$

The height loop reads altitude and vertical velocity and the denominator compensates tilt, the guard 0.35 limiting compensation to about $69.5°$; the upper clip does not supersede rotor limits. The CMG path has no direct collective output but reaches height through attitude, tilt compensation, and allocation saturation, so the shared interface isolates changes in requested attitude torque.

III. CMG State Dynamics and Online Decoder Adaptation

This section specifies the residual architecture in state-space form and the online decoder update, then states the boundedness properties that follow from the resulting saturations and projections.

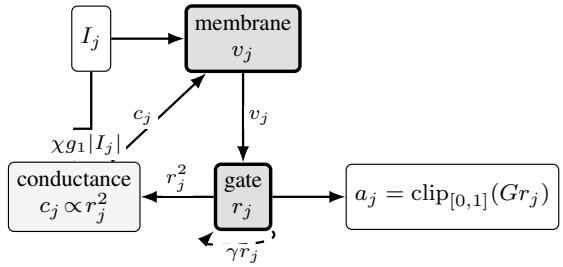


Fig. 2. Coupling topology of one CMG unit, cf. (19). The membrane sets the gate target; the gate acts back twice, shifting its own threshold through γ and shunting the membrane through $c_j \propto r_j^2$ ($\chi = 1$ makes that shunt input dependent). The dynamics are continuous: no spike, no reset, and only a_j leaves the unit.

A. Encoding, States, and Numerical Update

For m torque channels and $2m$ units, normalize the clipped baseline and form positive/negative current pairs:

$$\begin{aligned} \bar{\boldsymbol{u}} &= \boldsymbol{u}_0 \oslash \boldsymbol{\ell}, \\ \boldsymbol{I}^\pm &= I_b \mathbf{1} + I_g \max(\pm \bar{\boldsymbol{u}}, \mathbf{0}), \\ \boldsymbol{I} &= [(\boldsymbol{I}^+)^\mathsf{T}, (\boldsymbol{I}^-)^\mathsf{T}]^\mathsf{T}. \end{aligned} \tag{16}$$

Each unit has a dimensionless membrane state v_j and gate r_j:

$$\dot{r}_j = [\sigma(\alpha_h(v_j - v_{\mathrm{th}} + \gamma r_j)) - r_j]/\tau_r, \tag{17}$$

$$\dot{v}_j = -(v_j - v_{\mathrm{rest}})/\tau + I_j/\tau - c_j(v_j - v_{\mathrm{reset}}), \tag{18}$$

$$c_j = (g_0 + \chi g_1 |I_j|) r_j^2, \qquad \chi \in \{0, 1\}. \tag{19}$$

The logistic gate target depends on v_j, while r_j shifts that target and modulates membrane conductance (Fig. 2). Thus, both states participate even though the decoder reads the gate-derived activity. The model has no threshold-triggered reset or spike event; v_{reset} is a reversal level in a continuous conductance term. The gains g_0, g_1 have the units needed to make c_j a rate.

The actual integration is sequential, not exponential Euler for both states. From the old states compute

$$b_{j,k} = \sigma\Big(\mathrm{clip}_{[-60,60]}[\alpha_h(v_{j,k} - v_{\mathrm{th}} + \gamma r_{j,k})]\Big), \tag{20}$$

$$r^+_{j,k} = \mathrm{clip}_{[0,1]}[r_{j,k} + (h/\tau_r)(b_{j,k} - r_{j,k})]. \tag{21}$$

Then evaluate $c_{j,k} = (g_0 + \chi g_1 |I_{j,k}|)(r^+_{j,k})^2$, $\lambda_{j,k} = \tau^{-1} + c_{j,k}$, and $d_{j,k} = (v_{\mathrm{rest}} + I_{j,k})/\tau + c_{j,k} v_{\mathrm{reset}}$. The frozen-coefficient membrane step is

$$v^+_{j,k} = e^{-\lambda_{j,k}h} v_{j,k} + (1 - e^{-\lambda_{j,k}h}) d_{j,k}/\lambda_{j,k}. \tag{22}$$

This exactly solves the scalar membrane equation with those frozen coefficients; it does not establish unconditional stability of the coupled nonlinear closed loop. Assign $(v_{j,k+1}, r_{j,k+1}) = (v^+_{j,k}, r^+_{j,k})$. The exponential-Euler step $r^+_{j,k} = e^{-h/\tau_r} r_{j,k} + (1 - e^{-h/\tau_r}) b_{j,k}$, exact for a frozen target, is convex for every $h > 0$ without a clip [27]; it changes only the weight on $r_{j,k}$ (0.905 vs. 0.9 at $h/\tau_r = 0.1$), and all results use (21).

B. Readout, PES, and Additive Residual

The within-step activity, initial decoder, and clipped readout are

$$\begin{aligned} \boldsymbol{a}_k &= \mathrm{clip}_{[0,1]}(G \boldsymbol{r}^+_k), \qquad W_0 = [I_m, -I_m], \\ \boldsymbol{s}_k &= \mathrm{clip}_{[-1,1]}(W_k \boldsymbol{a}_k). \end{aligned} \tag{23}$$

Pair subtraction retains directional information from a nonnegative activity basis; state memory does not imply instantaneous sign agreement after a command reversal. With $G = 10^5$ the activity saturates once $r_j \geq 10^{-5}$, so G is a primary design hyperparameter for how much of r_j's dynamic range reaches the decoder. The teaching target and local update are

$$\boldsymbol{y}^*_k = \mathrm{clip}_{[-1,1]}(-k_e \boldsymbol{e}_{q,k} - k^*_\omega \boldsymbol{\omega}_k), \tag{24}$$

$$\boldsymbol{\varepsilon}_k = \boldsymbol{y}^*_k - \boldsymbol{s}_k, \tag{25}$$

$$W_{k+1} = \Pi_{\mathcal{W}}[W_k + \eta \boldsymbol{\varepsilon}_k \boldsymbol{a}_k^{\mathsf{T}} - \lambda_W W_k]. \tag{26}$$

Here $\mathcal{W} = [-W_{\max}, W_{\max}]^{m \times 2m}$. The coefficients η and λ_W are per-step (so changing h changes the effective learning dynamics). PES is a local error-driven update toward the prescribed target $\boldsymbol{y}^*_k$; it does not differentiate through the plant or claim to minimize a closed-loop cost. Since $\boldsymbol{s}_k$ is normalized by the authority $\beta\boldsymbol{\ell}$, anisotropic scaling enters only through $\boldsymbol{\ell}$ and one pair (k_e, k^*_ω) serves all axes.

Using W_k before the update, the requested command is

$$\boldsymbol{u}_k = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_{0,k} + \beta \boldsymbol{\ell} \odot \boldsymbol{s}_k), \quad \boldsymbol{\delta}_k = \boldsymbol{u}_k - \boldsymbol{u}_{0,k}, \quad 0 \leq \beta \leq 1. \tag{27}$$

The canonical UAV case has $m = 3$, six CMG units, and eighteen decoder weights. Intrinsic conductance modulation (χ), synaptic/decoder adaptation (W_k), and output authority ($\beta\boldsymbol{\ell}$) are distinct.

C. Structural Properties and Their Scope

We first state the boundedness properties the architecture gives directly: internal-state bounds, a command-deviation bound set by residual authority, and projection bounds on the decoder.

Proposition 1 (Bounded internal states)**.** *With the nonnegative conductance parameters in Table I and $r_{j,0} \in [0,1]$, the gate remains in $[0,1]$ for every $h > 0$, and the gate clip in* (21) *is inactive when $0 < h \leq \tau_r$. The membrane iterate remains in the convex hull of its initial value, v_{reset}, and the interval $[v_{\mathrm{rest}} + I_b, v_{\mathrm{rest}} + I_b + I_g]$.*

Proof. For $0 < h \leq \tau_r$ the unclipped gate step is a convex combination of $r_{j,k}$ and $b_{j,k} \in [0,1]$, so the clip is inactive; for $h > \tau_r$ it enforces the bound. Since $I_j \in [I_b, I_b + I_g]$ and $c_j \geq 0$, both d_j/λ_j and the update (22) are convex combinations of the stated endpoints; induction closes both claims. □

Proposition 2 (Bounded authority and baseline recovery)**.** *For $\boldsymbol{u}_{0,k} \in \mathcal{U}$ and any finite W_k, $\boldsymbol{u}_k \in \mathcal{U}$ and $|\delta_{k,i}| \leq \beta\ell_i$. If $\beta = 0$, the requested baseline command is recovered at every step. If $W_k = 0$ at a particular step, it is recovered at that step, but learning need not keep $W_{k+1} = 0$.*

Proof. The outer clip enforces membership in $\mathcal{U}$; its componentwise nonexpansiveness with $\mathrm{sat}_{\ell_i}(u_{0,k,i}) = u_{0,k,i}$ gives $|\delta_{k,i}| \leq |\beta \ell_i s_{k,i}| \leq \beta\ell_i$, and the recovery statements follow by substitution. □

The proposition establishes neither state safety, rotor headroom, stability, nor a bound on tracking improvement; a bounded adversarial correction can still degrade or destabilize a nonlinear plant.

Proposition 3 (Decoder bound and restricted contraction)**.** *If $W_0 \in \mathcal{W}$, then* (26) *keeps $W_k \in \mathcal{W}$. If, additionally, the target and a nonzero activity $\boldsymbol{a}$ are fixed, leak is zero, and readout clipping and projection remain inactive, then*

$$\boldsymbol{\varepsilon}_{k+1} = (1 - \eta \|\boldsymbol{a}\|_2^2) \boldsymbol{\varepsilon}_k. \tag{28}$$

For a nonzero initial error, its norm contracts to zero when $0 < \eta \|\boldsymbol{a}\|_2^2 < 2$.

Proof. Projection gives the first result; under the additional conditions $(W_{k+1} - W_k)\boldsymbol{a} = \eta \boldsymbol{\varepsilon}_k \|\boldsymbol{a}\|_2^2$, and subtracting it from the fixed target error gives (28), whose scalar multiplier has modulus below one exactly as stated. □

The readout clip is a nonexpansive projection onto a convex set containing $\boldsymbol{y}^*_k$ [28], so saturation never enlarges the PES error. This is the constant-feature PES result [11], not a closed-loop convergence theorem: with $\|\boldsymbol{a}\|^2 \leq 2m$, $0 < \eta < 1/m$ suffices in that restricted model, while changing targets, loss of excitation, clipping, and leakage invalidate a direct application to the experiment, and one fixed regressor identifies no unique decoder.

D. Allocation-Aware Local Stability

Let $E = [\mathbf{0}_{3\times1}, I_3]^\mathsf{T}$, $H = E^\mathsf{T}$, and let $\mathcal{A}_T(\boldsymbol{u}) = HB\,\mathrm{clip}_{[0,\boldsymbol{f}_{\max}]}(B^{-1}[T,\boldsymbol{u}^\mathsf{T}]^\mathsf{T})$ be the realized torque at collective T. For two commands at the *same state and collective*, scalar clipping has secant slopes $d_i \in [0,1]$. Consequently,

$$\mathcal{A}_T(\boldsymbol{u}) - \mathcal{A}_T(\boldsymbol{u}_0) = HBDB^{-1}E\boldsymbol{\delta}, \tag{29}$$

$$\|\mathcal{A}_T(\boldsymbol{u}) - \mathcal{A}_T(\boldsymbol{u}_0)\|_2 \le L_A\beta\|\boldsymbol{\ell}\|_2, \tag{30}$$

where $D = \mathrm{diag}(d_i)$ and $L_A = \max_{D\in\mathcal{D}}\|HBDB^{-1}E\|_2$. $\mathcal{D}$ is the set of sixteen diagonal matrices with entries in $\{0,1\}$. The vertex maximum bounds all intermediate D by convexity. Numerically $L_A = 4.53$ (X2) and 3.33 (CF2), inflated because small $|\kappa_i|$ turn clipped yaw requests into roll/pitch torque; if all of $\mathcal{U}$ is rotor-feasible at the operating collective (CF2 at hover, not X2), $D = I$ and 1 replaces L_A. The bound holds under rotor clipping but relates two commands at one state, not two closed-loop trajectories, and does not bound the height error.

For a constant reference and a smooth shortest-path branch, let $\boldsymbol{x} = [\boldsymbol{e}_q^\mathsf{T}, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T}$ and $K_q = \mathrm{diag}(\boldsymbol{k}_q)$, $K_\omega = \mathrm{diag}(\boldsymbol{k}_\omega)$. Write the continuous-time attitude subsystem as

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + \boldsymbol{g}(\boldsymbol{x}) + F\boldsymbol{d},$$
$$A = \begin{bmatrix} 0 & \frac{1}{2}I_3 \\ -\boldsymbol{J}^{-1}K_q & -\boldsymbol{J}^{-1}K_\omega \end{bmatrix}, \qquad F = \begin{bmatrix} 0 \\ \boldsymbol{J}^{-1} \end{bmatrix}, \tag{31}$$

where $\boldsymbol{d} = \boldsymbol{\tau}_{\mathrm{act}} + \boldsymbol{\tau}_{\mathrm{p}} + \boldsymbol{\tau}_{\mathrm{d}} + K_q\boldsymbol{e}_q + K_\omega\boldsymbol{\omega}$ is the total deviation from ideal PD torque. It includes allocation and command-saturation errors, not just the neural request. The nonlinear remainder is

$$\boldsymbol{g}(\boldsymbol{x}) = \begin{bmatrix} \frac{1}{2}[(\sqrt{1-\|\boldsymbol{e}_q\|^2}-1)I_3 + [\boldsymbol{e}_q]_\times]\boldsymbol{\omega} \\ -\boldsymbol{J}^{-1}[\boldsymbol{\omega}]_\times\boldsymbol{J}\boldsymbol{\omega} \end{bmatrix} = O(\|\boldsymbol{x}\|^2), \tag{32}$$

with $[\boldsymbol{z}]_\times\boldsymbol{y} = \boldsymbol{z}\times\boldsymbol{y}$. Norms below use fixed coordinates with angular rates expressed in rad/s. The first block's two terms are orthogonal and $(1-\sqrt{1-\|\boldsymbol{e}_q\|^2})^2 \le \|\boldsymbol{e}_q\|^2$, so it is at most $\frac{\sqrt{2}}{2}\|\boldsymbol{e}_q\|\|\boldsymbol{\omega}\|$; since $\boldsymbol{\omega}\times\boldsymbol{J}\boldsymbol{\omega} = \boldsymbol{\omega}\times(\boldsymbol{J}-\mu I_3)\boldsymbol{\omega}$, the choice $\mu = \frac{1}{2}(\lambda_{\max}+\lambda_{\min})(\boldsymbol{J})$ bounds the second by $\frac{1}{2}(\kappa(\boldsymbol{J})-1)\|\boldsymbol{\omega}\|^2$, with κ the condition number. With $\|\boldsymbol{e}_q\|\|\boldsymbol{\omega}\| \le \frac{1}{2}\|\boldsymbol{x}\|^2$,

$$\|\boldsymbol{g}(\boldsymbol{x})\| \le L_r\|\boldsymbol{x}\|^2, \qquad L_r = \tfrac{\sqrt{2}}{4} + \tfrac{1}{2}(\kappa(\boldsymbol{J})-1), \tag{33}$$

i.e., $L_r = 1.048$ (X2) and 0.529 (CF2), above sampled suprema 0.483 and 0.283; the cruder $\sqrt{2}/4+\kappa(\boldsymbol{J})$ also holds.

Theorem 1 (Conditional local practical stability)**.** *Let $\boldsymbol{J}, K_q, K_\omega \succ 0$, let q_d be constant, and use the smooth branch in* (31)*. Choose $Q_L \succ 0$ and the solution $P \succ 0$ of $A^\mathsf{T}P + PA = -Q_L$. On $\|\boldsymbol{x}\| \le r < 1$, suppose $\|\boldsymbol{g}(\boldsymbol{x})\| \le L_r\|\boldsymbol{x}\|^2$ (e.g.,* (33)*) and choose r so that $c = \lambda_{\min}(Q_L) - 2\|P\|L_r r > 0$. Set $p_- = \lambda_{\min}(P)$, $p_+ = \lambda_{\max}(P)$, $b = \|PF\|$, and $V = \boldsymbol{x}^\mathsf{T}P\boldsymbol{x}$. If*

$$V(0) \le p_- r^2, \qquad \|\boldsymbol{d}(t)\| \le \bar{d} < \frac{cr}{2b}\sqrt{\frac{p_-}{p_+}}, \tag{34}$$

then $V \le p_- r^2$ is forward invariant and

$$V(t) \le e^{-\alpha t}V(0) + \frac{4p_+b^2}{c^2}\bar{d}^2, \qquad \alpha = \frac{c}{2p_+}. \tag{35}$$

Thus attitude regulation is locally input-to-state stable; zero perturbation gives local exponential convergence of the plant state.

Proof. Along $\dot{\boldsymbol{x}} = A\boldsymbol{x}$ the mechanical energy $V_L = \boldsymbol{e}_q^\mathsf{T}K_q\boldsymbol{e}_q + \frac{1}{2}\boldsymbol{\omega}^\mathsf{T}\boldsymbol{J}\boldsymbol{\omega}$ obeys $\dot{V}_L = -\boldsymbol{\omega}^\mathsf{T}K_\omega\boldsymbol{\omega} \le 0$ (K_q symmetric, so products of inertia do not enter); on $\dot{V}_L \equiv 0$, $\boldsymbol{\omega} \equiv \mathbf{0}$ forces $\boldsymbol{e}_q = \mathbf{0}$, so LaSalle [29] makes A Hurwitz and $P \succ 0$ unique. Then (31)–(32) give $\dot{V} \le -c\|\boldsymbol{x}\|^2 + 2b\|\boldsymbol{x}\|\bar{d} \le -\frac{c}{2}\|\boldsymbol{x}\|^2 + \frac{2b^2}{c}\bar{d}^2$, and on $V = p_-r^2$, $\|\boldsymbol{x}\| \ge r\sqrt{p_-/p_+}$, so (34) makes the first bound strictly negative and keeps the ellipsoid inside the radius-r ball; integrating $\dot{V} \le -\alpha V + 2b^2\bar{d}^2/c$ gives (35). □

If the same-state baseline deviation from ideal PD is bounded by $\bar{d}_0$, (30) permits $\bar{d} = \bar{d}_0 + L_A\beta\|\boldsymbol{\ell}\|$ in the theorem whenever (34) holds: a sufficient local robustness condition independent of the PES trajectory, not a convergence proof. Maximizing that bound over r gives $r^* = \min\{\lambda_{\min}(Q_L)/(4\|P\|L_r), 1\}$ and, when $r^* < 1$,

$$\bar{d}_{\max} = \frac{\lambda_{\min}^2(Q_L)}{16\,b\,\|P\|\,L_r}\sqrt{\frac{p_-}{p_+}}, \qquad \beta < \beta_{\max} = \frac{\bar{d}_{\max} - \bar{d}_0}{L_A\|\boldsymbol{\ell}\|_2}, \tag{36}$$

a sufficient authority bound when $\bar{d}_0 < \bar{d}_{\max}$. With $Q_L = I$, A has spectral abscissa -1.395 (X2) and -3.325 (CF2), and the PD command is unsaturated and rotor-feasible in the certified ball, so $\bar{d}_0 = 0$ at hover and $(r^*, \bar{d}_{\max}, \beta_{\max}) = (5.31\times10^{-2},\ 3.90\times10^{-4}\,\mathrm{N\,m},\ 9.69\times10^{-5})$ for X2, $(1.05\times10^{-1},\ 1.11\times10^{-6}\,\mathrm{N\,m},\ 6.43\times10^{-4})$ for CF2 (2.14×10^{-3} with exact allocation). The experimental $\beta = 0.5$ lies far outside this conservative certificate: stability of the reported runs is observed, not certified. Moving references, the 180° sign boundary, sampled-data error, and translational stability remain outside the result.

Remark 1 (Sign selection on certified trajectories)**.** *Under Theorem 1, $\|\boldsymbol{e}_q\| \le \|\boldsymbol{x}\| \le r < 1$ gives $|q_{e,0}| \ge \sqrt{1-r^2} > 0$: sign selection in* (6) *never switches, the smooth branch is self-consistent, and $q_{e,0}$ perturbations below $\sqrt{1-r^2}$ cannot flip it; the global obstruction [19] remains.*

a) Nengo realization.: In Nengo 4.1.0 a custom `NeuronType` exposes membrane and gate states and direct current injection bypasses NEF gain/intercept fitting; unfiltered CMG activity uses $\eta_N = \eta(2m)/h$ and error $\boldsymbol{s} - \boldsymbol{y}^*$, with leak on pre-step weights. Its teaching graph adds a one-step latency, so weight sequences differ; over ten direct/Nengo cases the maximum relative mean-error difference is 2.013×10^{-6}: tested numerical agreement, not formal equivalence.

IV. Experimental Protocol and Cross-Mission Results

A. Models, Parameters, and Measurements

MuJoCo [30] simulates two Menagerie-derived multirotor models [31], CF2 and X2, as free-flying rigid bodies with site-applied wrenches; the attitude loop commands three body torques and a height loop the collective thrust.

TABLE I

MODEL/INTERFACE PARAMETERS WITH MANUFACTURER CONTEXT (TOP) AND CANONICAL 3-CHANNEL CMG–PES PARAMETERS (BOTTOM). MASS AGREES WITH PUBLISHED CF2/X2 SPECIFICATIONS [33], [34]; FORCE LIMITS, INERTIAS, AND REACTION COEFFICIENTS ARE SIMULATION PARAMETERS, NOT IDENTIFIED HARDWARE RATINGS. NEURAL STATES/CURRENTS ARE NORMALIZED MODEL VARIABLES; LEARNING COEFFICIENTS ARE PER STEP.

Parameter	CF2 rotor model	X2 model
Mass M (kg)	0.027	1.325
Rotor $\lvert p_x\rvert, \lvert p_y\rvert$ (m)	$0.0325, 0.0325$	$0.14, 0.18$
Reaction $\lvert\kappa_i\rvert$ (m)	0.005	0.0201
Rotor/total force limit (N)	$.0875 \to .13125$ / $.350 \to .525$	13 / 52
Main-test thrust/weight	1.982	4.001
$\boldsymbol{j}_{\mathrm{tune}}$ (kg m^2)	$10^{-5}(2.3951, 2.3951, 3.2347)$	$10^{-2}(6.0711, 3.6468, 2.5412)$
Integrator, timestep	RK4, 0.0005 s	Euler, 0.0005 s
Mfr. thrust data	.568 N[a]	Not published[b]

[a]Historical CF2 measurement at 93.75% PWM on an external supply [35], not a continuous battery rating; the modeled .525 N is below it. [b]Not in the reviewed X2 datasheet. Neither model includes motor lag; a historical CF2 motor step took about 180 ms [36].

CMG–PES	Value		Value
h, duration	0.0005 s, 45 s	$\alpha_h, \gamma, g_0, g_1$	$50, 2, 120, 30$
τ, τ_r	0.05 s, 0.005 s	I_b, I_g, G	$0.75, 1.25, 10^5$
$v_{\mathrm{rest}}, v_{\mathrm{reset}}, v_{\mathrm{th}}$	$0, 0, 1$	β, η	$0.5, 8 \times 10^{-4}$
$v_{j,0}, r_{j,0}, \chi$	$0, 0, 1$	$\lambda_W, W_{\max}, k_e, k_\omega^*$	$10^{-6}, 6, 8, 0$

The upstream CF2 model exposes abstract moment channels [32]; we replace them with four rotor-force channels and the clipped inverse allocator (14), ordering rotors $(-x,-y),(-x,+y),(+x,+y),(+x,-y)$ with reaction signs $(-,+,-,+)$. The CF2 $1.5\times$ scale (Table I) raises a model limit, not measured hardware capability. Both models omit motor/RPM/battery states and validated rotor aerodynamics; feedback uses `qpos`/`qvel`, with noise only in Sec. IV-D.

CF2's body axes are principal axes; X2's geometry-derived body tensor, $\boldsymbol{J} = 10^{-2}[3.66517, 0, -.21000;\ 0, 2.54117, 0;\ -.21000, 0, 6.05280]\,\mathrm{kg\,m^2}$, differs from both $\mathrm{diag}(\boldsymbol{j}_{\mathrm{tune}})$ and a manufacturer-identified tensor. Torque limits use $\boldsymbol{j}_{\mathrm{tune}}$.

Each PD/CMG pair shares initial state, reference, duration, allocator, model, and timestep. The primary campaign is nominal and deterministic (no injected disturbances or sensor noise), one run per cell, learning active throughout. We report the mean error $\bar{\theta}_q$ and the reduction $100(\bar{\theta}_{q,\mathrm{PD}} - \bar{\theta}_{q,\mathrm{CMG}})/\bar{\theta}_{q,\mathrm{PD}}$ over all 90,000 samples, even when plotted traces are decimated.

The five reference triples are generated as follows (radians): Roll is $(0.25\sin 0.7t, 0, 0)$; Full3 is $0.5\sin(t + (0, \pi/2, \pi)^{\mathsf{T}})$, componentwise; Hard45 is $(0.55\sin 0.75t + 0.20\sin(2.10t{+}0.40),\ 0.45\sin(0.95t{+}\pi/3){+}0.18\sin 1.75t,\ 0.65\sin(0.55t + \pi) + 0.25\sin(1.35t + 0.20))$. Hardstep holds zero for one second, then cycles every two seconds through $(.75, -.45, .55)$, $(-.65, .55, -.70)$, $(.50, .50, .90)$, and $(-.80, -.35, .20)$. Yaw360 is $(0, 0, 2\pi(3\xi^2 - 2\xi^3))$ with $\xi = t/45$. Hover is omitted: its near-zero error admits no meaningful percentage.

B. Primary Five-Mission Matrix

CMG-PES improves mean quaternion-angle tracking error in all ten nominal missions while respecting the same actuator interface as the PD baseline. The largest reductions occur on Yaw360 and Full3 and the smallest on Roll. Hardstep has the largest absolute error and the only nonzero rotor-bound occupancy S (samples with a rotor command at a bound, PD/CMG): $.016/.239$ on X2 and $.137/.264$ on CF2, a post-clipping statistic rather than a measure of requested-wrench infeasibility. These cells establish deterministic nominal performance relative to (11), not optimality over classical controllers; Sec. IV-C quantifies that gap. Fig. 3 pairs X2 Full3 with its internal CMG signals, and improvement refers to the mean metric, not every instant.

TABLE II

CANONICAL NENGO CMG–PES VERSUS PD: MEAN QUATERNION-ANGLE ERROR [RAD] AND REDUCTION VS PD [%], 45 S, $n = 1$ PER CLEAN CELL. FIXED: SAME UNITS WITH $W_k = W_0$; MOD./REAL.: MODERATE (0.010 RAD, 0.025 RAD S^{-1}) AND REALISTIC (0.020 RAD, 0.050 RAD S^{-1}) NOISE OF SEC. IV-D, TWO-SEED MEANS, SEED SD $\leq$ 0.03 PP. CF2 IS THE $1.5\times$ ROTOR-AUTHORITY DESIGN, NOT STOCK HARDWARE.

		Error [rad]		Reduction vs PD [%]			
Model	Mission	PD	CMG–PES	PES	Fixed	Mod.	Real.
X2	Roll	.050565	.041785	17.36	1.93	32.3	32.2
	Full3	.303275	.131748	56.56	9.74	56.4	55.7
	Hard45	.294082	.138007	53.07	10.27	52.9	52.3
	Hardstep	.454528	.359355	20.94	0.46	20.8	20.6
	Yaw360	.075863	.029902	60.58	0.12	56.8	47.8
CF2	Roll	.051337	.042439	17.33	3.31	32.4	32.5
	Full3	.288383	.126476	56.14	4.53	55.9	55.2
	Hard45	.285449	.132319	53.65	6.80	53.4	52.7
	Hardstep	.426182	.347886	18.37	0.42	18.3	18.1
	Yaw360	.075812	.029870	60.60	0.06	56.8	47.8

C. Baseline Ladder: Feedforward and Bandwidth

Equation (11) damps absolute rate and so carries a velocity lag on a moving reference. Table III repeats the ten cells on (12), at the Table II gains and at the swept optimum; the second reading below is unfavourable to the architecture.

(i) The residual improves every rung, by a shrinking margin. Geometric means are $44.5/44.2\%$ over PD (X2/CF2), $57.8/57.6\%$ over PD+ff at $\rho = 1$, and $35.6/6.5\%$ at $\rho = 3$; the best controller here is CMG–PES on the tuned baseline, on both models. *(ii) A tuned feedforward PD alone outperforms the augmented controller of Table II.* Feedforward at unchanged gains removes 76–99% of PD's error on the moving-reference missions, and at $\rho = 3$ the classical law wins every cell: by 13–$300\times$ on the smooth missions and .266 versus .359 rad on X2 Hardstep. The 17.3–60.6% of Table II is a statement about (11), not about the best classical law of this family. *(iii) Hardstep inverts at high bandwidth.* Its reference is piecewise constant, so feedforward changes nothing ($-.1/ + .1\%$) and only bandwidth helps ($41.5/38.4\%$); there the residual *degrades* tracking by 17.0% (X2) and 1.3% (CF2).

Bandwidth was swept over $\rho \in [0.75, 8]$ on both models and both noise levels of Sec. IV-D; mission-mean error is

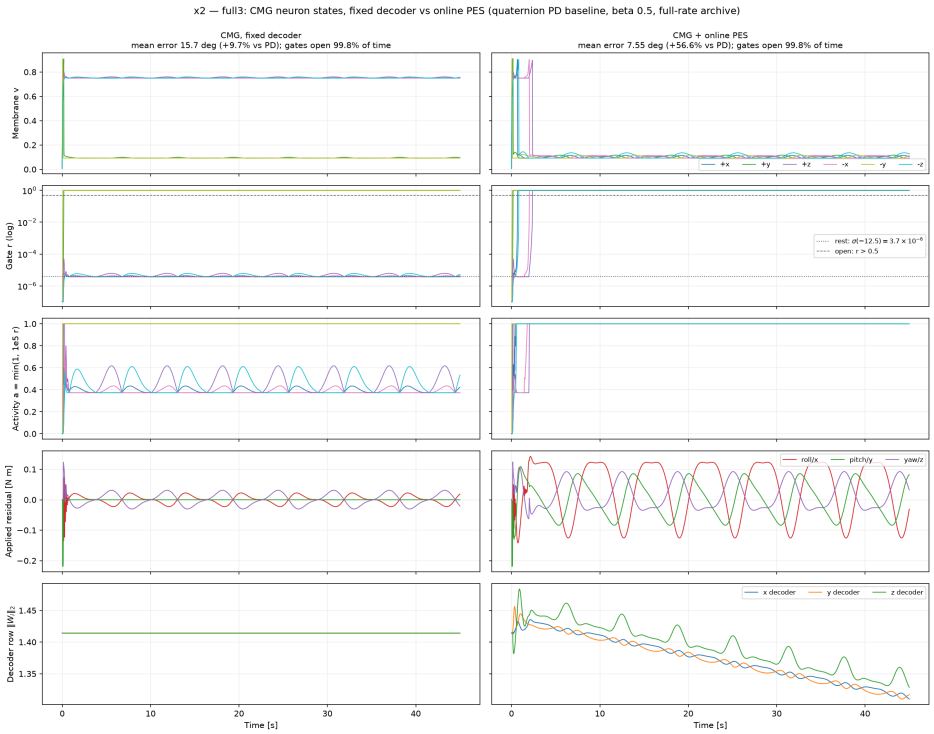


Fig. 3. X2 Full3: internal CMG signals on the 45-s trajectory, comparing fixed-decoder CMG (left) to online PES adaptation (right). Rows separate membrane state, raw gate r_j, and the clipped activity a_j presented to the decoder.

minimised at $\rho = 3$ in all four combinations (X2 noise-free .1205 $\rightarrow$ .0574, next best .0596 at $\rho = 4$), so the Table II gains sit a factor of three below the optimum of their own law, with rotor authority setting the ceiling: CF2 diverges from $\rho = 4$ (Hardstep 2.15 rad, a rotor at a bound in 83% of samples). Under a geometric mean the smooth missions keep improving past $\rho = 3$. Authority was selected per model by a rule fixed before these runs: X2 admits $\beta = .5$, CF2 only $\beta = .05$, since $\beta \geq .1$ diverges on CF2 Hardstep. Realized effort falls as the baseline improves (.121 $\rightarrow$ .0216 $\rightarrow$ .0148 N m residual RMS on X2 Full3), with zero saturation on every moving-reference cell. Sections IV-D–IV-G retain the PD baseline; restating them on (12) is future work.

D. Sensitivity to Measurement Noise

Each case was rerun at the two noise levels of Table II with two paired seeds: independent zero-mean Gaussian samples form a small Euler-angle quaternion left-multiplied into q and corrupt each rate component. These are not calibrated sensor specifications, and θ_q is evaluated from the *corrupted* quaternion the controller uses, so those columns measure noisy-feedback error. All noisy reductions stay positive, 18.08–55.75% at the higher level; Yaw360 loses about 12.8 pp against clean feedback while Roll gains. Seed SDs are .002–.031 pp (realistic): limited repeatability, not a confidence interval.

E. Ablation: What Drives the Improvement?

We separate decoder adaptation (PES), conductance modulation (χ), and readout gain (G); toggling χ, which retains the base shunt $g_0 r_j^2$, is numerically indistinguishable at a fixed decoder mode. *Fixed versus adaptive decoder.* Holding $W_k = W_0$ keeps the neuron dynamics but reduces the readout to $\boldsymbol{a}^+ - \boldsymbol{a}^-$, a bounded stateful image of the PD request. It still helps (X2 0.12–10.27%, CF2 0.06–6.80%), recovering at most a fifth of the PES gain in any cell; Fig. 3 shows why, row by row. *Membrane and gate.* At $I_j = I_b$ the membrane rests at $v_j \approx 0.75 < v_{\text{th}}$, so $r_j \approx 3.7 \times 10^{-6}$: the gate sits deep in its logistic tail, not closed. Once $r_j \geq 1/2$, $\gamma = 2$ and $v_j \geq 0$ (Proposition 1) give $b_{j,k} \geq 1/2$ in (21), so an opened gate cannot close (none did in twenty runs). *Activity.* With $G = 10^5$ the resting gate gives $a_j \approx 0.37$, graded by the input, and a latched gate pins $a_j = 1$. With

TABLE III

BASELINE LADDER ON THE SAME TEN CELLS: REFERENCE-RATE FEEDFORWARD (12) AT THE TABLE II BANDWIDTH AND AT THE SWEPT OPTIMUM $\rho = 3$, EACH WITHOUT AND WITH THE CMG–PES RESIDUAL. ERRORS IN 10^{-3} RAD; Δ IS THE RESIDUAL'S REDUCTION OVER THE BASELINE IN THE SAME BLOCK; GEO IS THE GEOMETRIC MEAN OVER THE FIVE MISSIONS.

		PD+ff, $\rho = 1$			PD+ff, $\rho = 3$ (tuned)		
Model	Mission	base	+CMG	Δ[%]	base	+CMG	Δ[%]
X2	Roll	2.98	0.61	79.5	0.34	0.074	78.4
	Full3	72.10	32.81	54.5	10.09	8.83	12.4
	Hard45	71.02	38.17	46.3	10.54	9.08	13.8
	Hardstep	454.77	359.16	21.0	265.91	311.07	-17.0
	Yaw360	1.78	0.60	66.4	0.20	0.12	41.8
	geo	26.19	11.04	57.8	4.54	2.93	35.6
CF2	Roll	4.91	0.97	80.2	0.56	0.43	22.9
	Full3	42.72	20.29	52.5	7.06	6.96	1.3
	Hard45	53.18	28.23	46.9	8.65	8.50	1.8
	Hardstep	425.93	347.68	18.4	262.59	265.93	-1.3
	Yaw360	0.74	0.25	66.4	0.084	0.079	5.6
	geo	20.39	8.65	57.6	3.77	3.52	6.5

the fixed decoder only the $\pm y$ pair latches on Full3, so $a_y^+ - a_y^- = 0$ and the pitch residual is exactly zero after 1 s while sub-threshold units carry roll and yaw. Under PES on X2 all gates latch by 2.4 s, and for constant $\bar{\boldsymbol{a}}$ with inactive projection $\boldsymbol{z}_k = W_k \bar{\boldsymbol{a}}$ obeys

$$\boldsymbol{z}_{k+1} = (1 - \lambda_W)\boldsymbol{z}_k + \eta \|\bar{\boldsymbol{a}}\|^2 [\boldsymbol{y}_k^* - \mathrm{clip}_{[-1,1]}(\boldsymbol{z}_k)], \quad (37)$$

the leaky constant-feature regime of Proposition 3: a first-order filter of the teaching target with pole 0.9952 (about 0.104 s) and DC gain $\approx$ 0.9998 while the readout clip is inactive. The latched residual is therefore *lagged proportional* action, not integral action, adding 1.50× (roll, pitch) and 1.63× (yaw) the PD stiffness on both vehicles behind a 9.60 rad/s corner. The smooth references are sums of constant-frequency components [37] whose largest instantaneous frequency, 2.1 rad/s (Hard45), lies below that corner (gain 0.977), while $\boldsymbol{e}_q$ noise above it is attenuated. *Residual and decoder.* The residual RMS rises from 0.026 to 0.123 N m and the decoder moves $\|W_T - W_0\|_F = 0.37$, with PES applying 4–12× the fixed residual RMS on both models. Where no gate opens (Roll, Yaw360; $r_j \leq 1.6 \times 10^{-3}$) PES lifts the driven membranes toward threshold and uses the graded tail. Decoder adaptation is therefore the principal lever: the CMG dynamics supply a bounded basis, latched or graded, on which PES shapes the correction. The leak λ_W is the σ-modification of robust adaptive control [38], leaving an exact PES mismatch of $2.1 \times 10^{-4}\|\boldsymbol{s}\|$ latched and $1.5 \times 10^{-3}\|\boldsymbol{s}\|$ at the tail.

F. Neuron-Family Comparison

The Nengo benchmark covers two models, five missions, and eleven controllers: PD and fixed/PES variants of CMG, LIF, ALIF, Izhikevich [39], and LIFRate, all sharing the bounded residual, saturation, and allocation interface; only the neuron dynamics and native-state-to-activity map differ. Table IV shows two patterns. First, PES is the dominant lever in every family: fixed decoders never approach their PES counterparts, even with more expressive dynamics, extending Sec. IV-E to every basis. Second, lower error is not free: Izhikevich–PES attains the largest mean reduction at three times the residual effort of CMG–PES, so equal output caps do not imply equal realized effort. Proposition 3 explains the flat rows: its factor $1 - \eta\|\boldsymbol{a}\|^2$ stays near one on a silent basis, and the LIF, ALIF and LIFRate bases are active in a few percent of unit-samples against $\geq$99.9% for CMG and Izhikevich. Persistent activity is necessary for PES here but does not set effort, since Izhikevich shares it.

G. Disturbance Rejection and Retention

An equilibrium test applies $\boldsymbol{\tau}_\mathrm{d} = d_b \varrho(t)[\sin(1.31t + .2), .8\sin(1.73t + 1.1), .6\sin(2.17t + 2)]^\mathsf{T}$ in body axes, ϱ alternating 5-s on/off intervals after 5 s of quiet, with $d_b = .15$ N m on X2. The four-burst 45-s run (Fig. 4) cuts burst-phase mean error from 11.50° to 6.65° (42.2%) and post-burst quiet error from 1.21° to .99° (18.0%); the CF2 rerun ($d_b = .00015$ N m) falls from .141199 to .070259 rad (50.24%).

TABLE IV

NENGO NEURON-FAMILY BENCHMARK, MEAN OVER TEN CELLS (TWO MODELS, FIVE MISSIONS): ERROR REDUCTION OVER THE SHARED PD BASELINE FOR A FIXED AND A PES DECODER, PES RESIDUAL EFFORT AS RMS FRACTION OF THE PER-AXIS LIMIT, AND THE FRACTION OF UNIT-SAMPLES WITH $|a_j| > 10^{-9}$.

Family	fixed [%]	PES [%]	PES effort	active
CMG	3.8	41.5	0.121	0.999
Izhikevich	2.6	46.8	0.371	1.000
LIF	0.5	1.6	0.036	0.028
ALIF	0.5	1.5	0.034	0.023
LIFRate	0.8	2.7	0.040	0.036

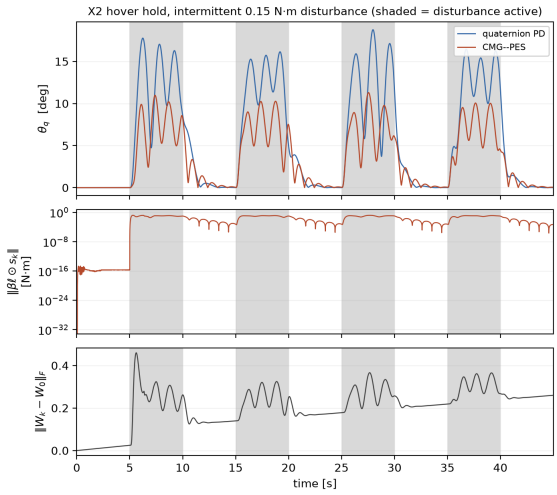


Fig. 4. X2 hover hold under four 0.15 N m disturbance bursts (shaded) separated by quiet intervals. Top: attitude error. Centre: residual magnitude, logarithmic. Bottom: decoder displacement from W_0. The residual is at numerical precision before the first burst. This means negligible initial correction, not zero computation or energy cost.

V. DISCUSSION AND CONCLUSION

A bounded residual of six continuous CMG units and eighteen PES-adapted decoder weights augments a fixed quaternion PD baseline without changing its gains or actuator interface, reducing mean quaternion-angle error by 17.3–60.6% over ten matched 45-s MuJoCo cases on two modeled platforms, within rotor saturation and with exact baseline recovery at zero authority; direct and Nengo realizations agree to 2.013×10^{-6} relative mean error. Sec. IV-C places that number: the residual also improves a feedforward PD (57.8/57.6%) and a bandwidth-tuned one (35.6/6.5%), yet the tuned classical law alone beats the PD-referenced augmented controller everywhere and the residual costs 17.0/1.3% on Hardstep there. It is complementary to a baseline missing a term, not a substitute for tuning one.

Decoder adaptation is the dominant source of improvement in the tested regime: a fixed decoder gives at most 10.3%, and toggling χ is indistinguishable. Opened gates latch and saturate the readout, where a fixed paired decoder can cancel an axis and PES reduces to lagged proportional action worth about 1.5–1.6× the PD stiffness, while sub-threshold units stay in a graded logistic tail. Across all five neuron families PES is the consistent differentiator, and the dynamics mainly set an effort–accuracy tradeoff. The

readout gain G, which sets the resting activity at 0.37, is a key sensitivity parameter, and the local certificate gives $\beta_{\max} \sim 10^{-4}$–$10^{-3}$, far below the tested $\beta = 0.5$.

Limitations: results are in simulation (Sec. IV-D bounds noise sensitivity; model mismatch and hardware effects are not addressed), the retention horizon is limited, and bounded commands, states, and weights are implementation properties, not stability proofs.

Future work includes restating the noise, ablation, neuron-family, and disturbance studies on the feedforward baseline, matched-effort comparisons against non-neural bounded bases, less conservative sampled-data certificates, held-out evaluations with frozen weights, 6-DOF extension, and validation on physical platforms; the same wrapper extends to nonlinear and optimal-control baselines with hard actuator constraints.

REFERENCES

[1] T. Johannink *et al.*, "Residual reinforcement learning for robot control," in *2019 International Conference on Robotics and Automation (ICRA)*, 2019, pp. 6023–6029.

[2] Z. Wu *et al.*, "L1Quad: L_1 adaptive augmentation of geometric control for agile quadrotors with performance guarantees," *IEEE Transactions on Control Systems Technology*, vol. 33, no. 2, pp. 597–612, 2025.

[3] X. Li, B.-C. Zheng, W. Wang, and W. Zhang, "HPA-PPO: A history-preview and attention-enhanced PPO for quadrotor attitude tracking," *Applied Mathematics and Computation*, vol. 532, p. 130236, 2027.

[4] I. Rubio Scola, L. R. Garcia Carrillo, and J. P. Hespanha, "Limbic system-inspired performance-guaranteed control for nonlinear multi-agent systems with uncertainties," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 8, pp. 4196–4207, 2023.

[5] J. Escareno, J. U. Alvarez-Munoz, L. R. Garcia Carrillo, I. Rubio Scola, J. Franco-Robles, and O. Labbani-Igbida, "Limbic system-inspired robust event-driven control for high-order uncertain nonlinear systems," *IEEE Control Systems Letters*, vol. 8, pp. 3057–3062, 2024.

[6] I. Rubio Scola, L. R. Garcia Carrillo, A. T. Sornborger, and J. P. Hespanha, "Biologically plausible robust control with neural network weight reset for unmanned aircraft systems under impulsive disturbances," in *2023 American Control Conference (ACC)*, 2023, pp. 4924–4929.

[7] I. Rubio Scola, G. A. Guijarro Reyes, L. R. Garcia Carrillo, J. P. Hespanha, and L. Burlion, "A robust control strategy with perturbation estimation for the Parrot Mambo platform," *IEEE Transactions on Control Systems Technology*, vol. 29, no. 4, pp. 1389–1404, 2021.

[8] C. Eliasmith and C. H. Anderson, *Neural Engineering: Computation, Representation, and Dynamics in Neurobiological Systems*. MIT Press, 2004.

[9] T. Bekolay *et al.*, "Nengo: A Python tool for building large-scale functional brain models," *Frontiers in Neuroinformatics*, vol. 7, p. 48, 2014.

[10] D. MacNeil and C. Eliasmith, "Fine-tuning and the stability of recurrent neural networks," *PLoS ONE*, vol. 6, no. 9, p. e22885, 2011.

[11] A. R. Voelker, "A solution to the dynamics of the prescribed error sensitivity learning rule," Centre for Theoretical Neuroscience, University of Waterloo, Tech. Rep. CTN-TR-20151001-01, Oct. 2015. [Online]. Available: https://www.compneuro.uwaterloo.ca/files/publications/voelker.2015.pdf

[12] T. DeWolf, T. C. Stewart, J.-J. Slotine, and C. Eliasmith, "A spiking neural model of adaptive arm control," *Proceedings of the Royal Society B: Biological Sciences*, vol. 283, no. 1843, p. 20162134, 2016.

[13] T. DeWolf, P. Jaworski, and C. Eliasmith, "Nengo and low-power AI hardware for robust, embedded neurorobotics," *Frontiers in Neurorobotics*, vol. 14, p. 568359, 2020.

[14] A. Olivares Cruz, E. S. Espinoza, L. E. Ramos Velasco, O. A. García Alcántara, L. R. Garcia Carrillo, and I. Rubio Scola, "Biologically-inspired self-tuning neuromorphic PID control for real-time trajectory tracking of a UAS," *IFAC-PapersOnLine*, vol. 59, no. 18, pp. 115–120, 2025.

[15] A. Olivares, E. S. Espinoza, L. E. Ramos, O. A. García Alcántara, L. R. Garcia Carrillo, and A. T. Sornborger, "Self-tuning neuromorphic controller for real-time UAS trajectory tracking based on prescribed error sensitivity," *Neural Computing and Applications*, vol. 38, no. 7, p. 192, 2026.

[16] A. L. Hodgkin and A. F. Huxley, "A quantitative description of membrane current and its application to conduction and excitation in nerve," *The Journal of Physiology*, vol. 117, no. 4, pp. 500–544, 1952.

[17] H. Schaub and J. L. Junkins, *Analytical Mechanics of Space Systems*, 4th ed. AIAA, 2018.

[18] J. Diebel, "Representing attitude: Euler angles, unit quaternions, and rotation vectors," *Matrix*, vol. 58, no. 15–16, pp. 1–35, 2006.

[19] S. P. Bhat and D. S. Bernstein, "A topological obstruction to continuous global stabilization of rotational motion and the unwinding phenomenon," *Systems & Control Letters*, vol. 39, no. 1, pp. 63–70, 2000.

[20] N. A. Chaturvedi, A. K. Sanyal, and N. H. McClamroch, "Rigid-body attitude control," *IEEE Control Systems Magazine*, vol. 31, no. 3, pp. 30–51, 2011.

[21] C. G. Mayhew, R. G. Sanfelice, and A. R. Teel, "Quaternion-based hybrid control for robust global attitude tracking," *IEEE Transactions on Automatic Control*, vol. 56, no. 11, pp. 2555–2566, 2011.

[22] I. Rubio Scola, O. A. García Alcántara, S. Sandoval, E. S. Espinoza Quesada, H. Haimovich, and L. R. Garcia Carrillo, "Globally asymptotically stable trajectory tracking of underactuated UAVs using geometric algebra," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 62, pp. 1491–1508, 2026.

[23] L. Escamilla, L. R. Garcia Carrillo, S. Sandoval, and E. S. Espinoza Quesada, "Stabilization and trajectory tracking of a subactuated aircraft based on a geometric algebra approach," in *2023 American Control Conference (ACC)*, 2023, pp. 3233–3238.

[24] E. Fresk and G. Nikolakopoulos, "Full quaternion based attitude control for a quadrotor," in *2013 European Control Conference (ECC)*, 2013, pp. 3864–3869.

[25] L. R. García Carrillo, A. E. Dzul López, R. Lozano, and C. Pégard, *Quad Rotorcraft Control: Vision-Based Hovering and Navigation*, ser. Advances in Industrial Control. London: Springer, 2013.

[26] J. C. Monteiro, F. Lizarralde, and L. Hsu, "Optimal control allocation of quadrotor UAVs subject to actuator constraints," in *2016 American Control Conference (ACC)*, 2016, pp. 500–505.

[27] M. Hochbruck and A. Ostermann, "Exponential integrators," *Acta Numerica*, vol. 19, pp. 209–286, 2010.

[28] H. H. Bauschke and P. L. Combettes, *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*, 2nd ed. Cham: Springer, 2017.

[29] H. K. Khalil, *Nonlinear Systems*, 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2002.

[30] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 5026–5033.

[31] Google DeepMind, "MuJoCo Menagerie," https://github.com/google-deepmind/mujoco_menagerie, model repository; accessed Sep. 5, 2026.

[32] ——, "Bitcraze Crazyflie 2 Description (MJCF)," https://github.com/google-deepmind/mujoco_menagerie/blob/main/bitcraze_crazyflie_2/README.md, model provenance and actuator-limit caveat; accessed Sep. 5, 2026.

[33] Bitcraze AB, "Crazyflie 2.0: Mechanical and Flight Specifications," https://store.bitcraze.io/products/crazyflie-2-0, accessed Sep. 5, 2026.

[34] Skydio, "Skydio X2E Datasheet," https://pages.skydio.com/rs/784-TUF-591/images/skydio-x2e-datasheet-x2-pg.pdf, aircraft specifications; accessed Sep. 5, 2026.

[35] Bitcraze AB, "PWM to Thrust: Old Measurements (2015)," https://www.bitcraze.io/documentation/repository/crazyflie-firmware/master/functional-areas/pwm-to-thrust/, historical test setup and total-thrust table; accessed Sep. 5, 2026.

[36] ——, "Measuring Propeller RPM: Part 3," https://www.bitcraze.io/2015/02/measuring-propeller-rpm-part-3/, Feb. 2015, published Feb. 9, 2015; accessed Sep. 5, 2026.

[37] S. Sandoval and P. L. De Leon, "The instantaneous spectrum: A general framework for time-frequency analysis," *IEEE Transactions on Signal Processing*, vol. 66, no. 21, pp. 5679–5693, 2018.

[38] P. A. Ioannou and J. Sun, *Robust Adaptive Control*. Upper Saddle River, NJ: Prentice Hall, 1996.

[39] E. M. Izhikevich, "Simple model of spiking neurons," *IEEE Transactions on Neural Networks*, vol. 14, no. 6, pp. 1569–1572, 2003.